

第二章 函数的导数(Derivative)

2.1 导数的定义(The Definition of Derivative)



Multiple Choice Questions

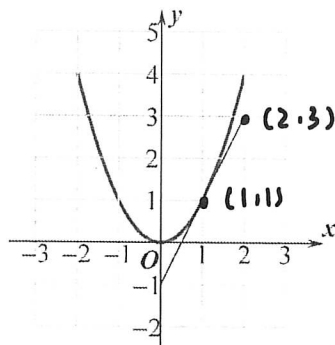
1. Which of the following values is the average rate of change of $f(x) = \sqrt{x+2}$ over the interval $[-1, 2]$?

(A) -4 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) 3

2. Let $f(x) = 4 - 3x$. Which of the following is equal to $f'(0)$?

(A) 4 (B) -4 (C) -3 (D) 4

3. The graph of f and its tangent line at $x=1$ is shown below, find $f'(1) = ()$.



(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2

4. Let $f(x) = kx^3 - 2x$, ($k \neq 0$). The average rate of $f(x)$ over the interval $[-2, -1]$ is 3, find k .

(A) $\frac{5}{7}$ (B) $\frac{1}{7}$ (C) $-\frac{1}{7}$ (D) $-\frac{5}{7}$

5. **Calculator** Let $y = \tan e^x - x^2 + 1$. Find $\frac{dy}{dx} \big|_{x=0.5}$

(A) -12.056 (B) 0 (C) 271.066 (D) 273.066

6. **Calculator** The graph of $y = \sqrt{x^2 + 1} - 3x$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.354 (B) -0.001 (C) -2.666 (D) 0

B 7. The population of insects at time t days is modeled by the function P with first derivative $P'(t) = 0.1t^2 + 10t + 120$. Which of the following statements is true?

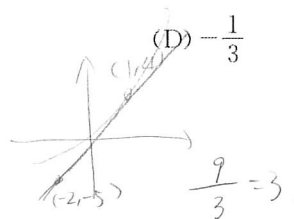
- (A) At time $t = 10$, the population of insects is 230.
 (B) At time $t = 10$, the population of insects is increasing at a rate of 230 insects per day.
 (C) At time $t = 10$, the population of insects is decreasing at a rate of 230 insects per day.
 (D) At time $t = 10$, the rate of change of the number of insects is increasing at a rate of 230 insects per day.

A 8. If the line tangent to the graph of a function g at the point $(1, 4)$ passes through the point $(-2, -5)$, what is the slope of this line?

- (A) 3 (B) -3 (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$



Free Response Questions



9. Let $f(x) = x^2 - 2x$

(a) Find the average rate of change of f over $[1, 3]$.

(b) Find $\frac{f(2+h) - f(2)}{h}$. ~~$2xh + h^2 - 4 - 2h + 2x$~~

(c) Find the instantaneous rate of change of f at $x = 2$.

$$\begin{aligned} (a) \quad & \frac{f(3) - f(1)}{3 - 1} \\ &= \frac{9 - 6 - (1 - 2)}{2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} (b) \quad & \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = \lim_{x \rightarrow 2} \frac{(x+h)^2 - 2(2+h) - x^2 + 2x}{h} \\ &= \frac{2xh + h^2 - 2h}{h} \\ &= 2x + h - 2 \\ &\because x = 2 \\ &= 4 + 0 - 2 = 2 \end{aligned}$$

(c) when $x = 2$

$$\begin{aligned} &= 4 + 0 - 2 = 2 \\ &= \frac{h^3 + 3xh^2 + 3x^2h}{h} \\ &= h^2 + 3xh + 3x^2 \end{aligned}$$

10. Use the definition of derivative to find the derivative of $f(x) = x^3$, then find $f'(2)$.

$$\begin{aligned} \because f(x) &= x^3 \\ \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} &= \frac{h^3 + 3xh^2 + 3x^2h}{h} \\ &= h^2 + 3xh + 3x^2 \end{aligned}$$

$$\because h \rightarrow 0$$

$$f'(x) = 3x^2$$

$$\therefore f'(2) = 3 \cdot 2^2 = 12$$